

CPU Scheduling Algorithms

Simulation Traces & Performance Metrics

CPU scheduling decides which process runs when there are multiple runnable processes. This guide uses a simulated CLI scheduler to visualize how FCFS, SJF, RR, and Priority algorithms manage process queues, Wait Time (WT), and Turnaround Time (TAT).

Sample Workload Data

To properly compare the algorithms, we will pass the following dataset (saved as `jobs.csv`) into our scheduling simulator. Priority assumes lower numbers equal higher priority.

```
user@os-sim:~/labs$ cat jobs.csv
Process, Arrival_Time, Burst_Time, Priority
P1, 0, 8, 3
P2, 1, 4, 1
P3, 2, 2, 4
P4, 3, 1, 2
```

1. FCFS (First-Come, First-Served)

The simplest scheduling algorithm. Processes are assigned to the CPU in the exact order they arrive. It is non-preemptive and can suffer from the "Convoy Effect" (short processes waiting for a long process to finish).

```
user@os-sim:~/labs$ ./scheduler --algo FCFS --file jobs.csv
Initializing FCFS Scheduler...

# Gantt Chart Execution Trace

[0]--P1--[8]--P2--[12]--P3--[14]--P4--[15]

# Process Metrics (TAT = Completion - Arrival, WT = TAT - Burst)

P1: Wait=0, Turnaround=8
P2: Wait=7, Turnaround=11
P3: Wait=10, Turnaround=12
P4: Wait=11, Turnaround=12

Average Wait Time: 7.00 ms
Average Turnaround Time: 10.75 ms
```

2. SJF (Shortest Job First - Non-Preemptive)

When the CPU is available, it is assigned to the process that has the smallest burst time. If a process is already running, it cannot be interrupted.

```
user@os-sim:~/labs$ ./scheduler --algo SJF_NP --file jobs.csv
```

```
Initializing SJF (Non-Preemptive) Scheduler...
```

```
# Gantt Chart Execution Trace
```

```
# At t=8, P2, P3, P4 are available. P4 has shortest burst (1).
```

```
[0]--P1--[8]--P4--[9]--P3--[11]--P2--[15]
```

```
# Process Metrics
```

```
P1: Wait=0, Turnaround=8
```

```
P2: Wait=10, Turnaround=14
```

```
P3: Wait=7, Turnaround=9
```

```
P4: Wait=5, Turnaround=6
```

```
Average Wait Time: 5.50 ms
```

```
Average Turnaround Time: 9.25 ms
```

3. SRTF (Shortest Remaining Time First)

The preemptive version of SJF. If a new process arrives with a burst time shorter than what is left of the currently executing process, the current process is interrupted (preempted).

```
user@os-sim:~/labs$ ./scheduler --algo SRTF --file jobs.csv
```

```
Initializing SRTF (Preemptive SJF) Scheduler...
```

```
# Gantt Chart Execution Trace
```

```
# P1 starts, but is preempted by P2 at t=1, P2 by P3 at t=2, P3 by P4 at t=3.
```

```
[0]-P1-[1]-P2-[2]-P3-[3]-P4-[4]-P3-[5]-P2-[8]-P1-[15]
```

```
# Process Metrics
```

```
P1: Wait=7, Turnaround=15
```

```
P2: Wait=3, Turnaround=7
```

```
P3: Wait=1, Turnaround=3
```

```
P4: Wait=0, Turnaround=1
```

```
Average Wait Time: 2.75 ms
```

```
Average Turnaround Time: 6.50 ms
```

4. RR (Round Robin)

Designed for time-sharing systems. Each process is given a small unit of CPU time (time quantum). If it doesn't finish, it is preempted and added to the end of the ready queue. Here, we use a Time Quantum of 3.

```
user@os-sim:~/labs$ ./scheduler --algo RR --quantum 3 --file jobs.csv
Initializing Round Robin Scheduler (Time Quantum = 3)...

# Gantt Chart Execution Trace

[0]--P1--[3]--P2--[6]--P3--[8]--P4--[9]--P1--[12]--P2--[13]--P1--[15]

# Process Metrics

P1: Wait=7, Turnaround=15
P2: Wait=8, Turnaround=12
P3: Wait=4, Turnaround=6
P4: Wait=5, Turnaround=6

Average Wait Time: 6.00 ms
Average Turnaround Time: 9.75 ms
```

5. Priority Scheduling (Non-Preemptive)

CPU is allocated to the process with the highest priority (lowest integer value = highest priority). If priorities are equal, FCFS rules apply. Note: Can cause "starvation" of low priority processes.

```
user@os-sim:~/labs$ ./scheduler --algo PRIORITY_NP --file jobs.csv
Initializing Priority (Non-Preemptive) Scheduler...
# Priority logic: 1 is highest, 4 is lowest.

# Gantt Chart Execution Trace
# P1 runs first (only arrival at t=0). At t=8, P2(Pri 1), P4(Pri 2), P3(Pri
4)

[0]--P1--[8]--P2--[12]--P4--[13]--P3--[15]

# Process Metrics

P1: Wait=0, Turnaround=8
P2: Wait=7, Turnaround=11
P3: Wait=11, Turnaround=13
P4: Wait=9, Turnaround=10

Average Wait Time: 6.75 ms
Average Turnaround Time: 10.50 ms
```